

Student Number: _____ Class Teacher: _____

St George Girls High School

Trial Higher School Certificate Examination

2016



Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen.
- Write your **student number** on each booklet.
- Board-approved calculators may be used.
- Marks may be deducted for careless or poorly presented work.
- A Reference Sheet is provided.
- In Question 11-16, show relevant mathematical reasoning and /or calculations

Total Marks – 100

Section I Pages 2 – 5

10 marks

- Attempt Questions 1 – 10.
- Allow about 15 minutes for this section.
- Answer on the multiple-choice sheet provided at the back of this paper.

Section II Pages 6 – 11

90 marks

- Attempt Questions 11 – 16.
- Allow about 2 hours 45 minutes for this section.
- Begin each question in a new booklet.

Section I	/10
Section II	
Question 11	/15
Question 12	/15
Question 13	/15
Question 14	/15
Question 15	/15
Question 16	/15
Total	/100

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1. Let $z = 2 - 3i$. What is the value of z^{-1} ?

(A) $-\frac{1}{5}(2 + 3i)$

(B) $\frac{1}{13}(2 + 3i)$

(C) $\frac{1}{5}(2 - 3i)$

(D) $\frac{1}{13}(2 - 3i)$

2. What is the value of $\int_0^2 \sqrt{\frac{x}{4-x}} dx$ using the substitution $x = 4 \sin^2 \theta$?

(A) 0.75π

(B) $\pi - 2$

(C) $\pi + 6$

(D) $3\pi - 8$

3. The polynomial $x^3 + 3x^2 + 2x - 1 = 0$ has roots α, β and γ .

Which polynomial has roots $\frac{2}{\alpha}, \frac{2}{\beta}$ and $\frac{2}{\gamma}$?

(A) $x^3 - 4x^2 - 12x - 8 = 0$

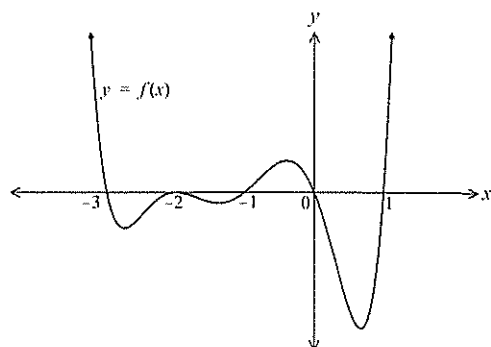
(B) $x^3 + 4x^2 - 12x + 8 = 0$

(C) $8x^3 - 12x^2 - 4x + 1 = 0$

(D) $8x^3 + 12x^2 + 4x - 1 = 0$

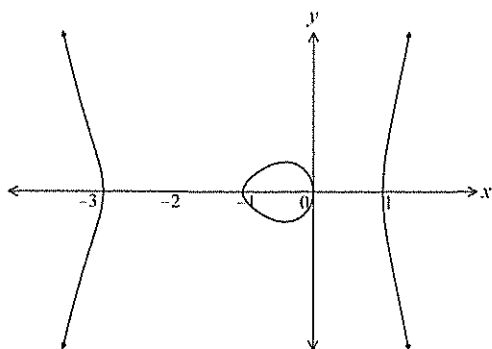
Section I (cont'd)

4. The diagram below shows the graph of the function $y = f(x)$.

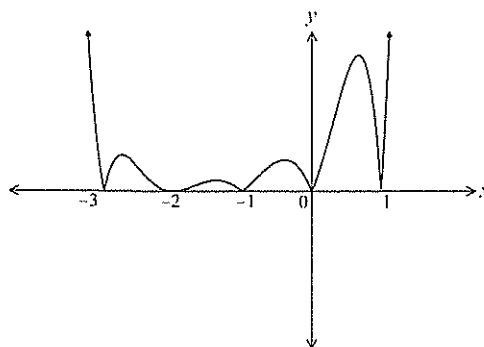


Which of the graphs below could represent the graph of $y = \frac{1}{f(x)}$?

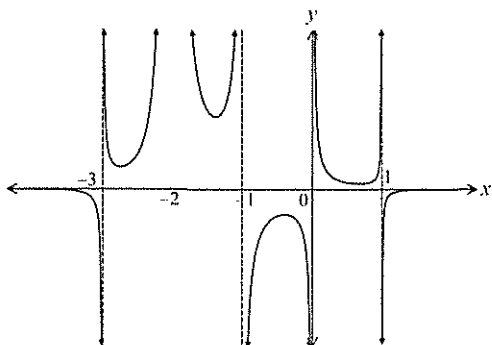
(A)



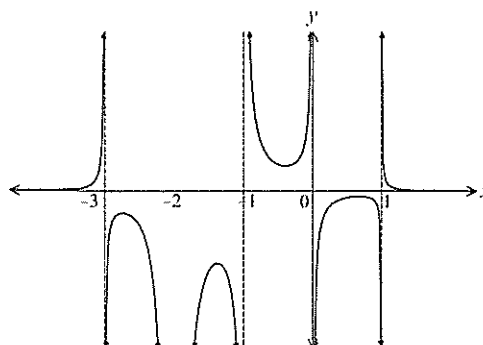
(B)



(C)



(D)



Section I (cont'd)

5. The area enclosed by the curve $y = 3x^2 - x^3$, the x -axis and the lines $x = 0$ and $x = 3$ is rotated about the y -axis.

What is the volume of the solid generated using the method of cylindrical shells?

- (A) $\frac{27\pi}{4}$
- (B) 12π
- (C) $\frac{243\pi}{10}$
- (D) $\frac{116\pi}{5}$
6. Given that $z = 3\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$, what is the value of $(\bar{z})^3$?
- (A) $9\left(\cos\frac{\pi}{2} - i\sin\frac{\pi}{2}\right)$
- (B) $9\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$
- (C) $27\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$
- (D) $27\left(\cos\frac{\pi}{2} - i\sin\frac{\pi}{2}\right)$

7. What is the value of $\int_2^3 \frac{1}{\sqrt{4x-x^2}} dx$?

- (A) $\frac{\pi}{2}$
- (B) $\frac{\pi}{3}$
- (C) $\frac{\pi}{6}$
- (D) $\frac{\pi}{4}$

Section I (cont'd)

8. Which expression gives the gradient of the normal to the curve $x^3 + xy + y^2 = 7$ at any point on the curve?
- (A) $\frac{-3x^2 - y}{x + 2y}$
- (B) $\frac{x + 2y}{3x^2 + y}$
- (C) $\frac{3x^2 + y}{x + 2y}$
- (D) $\frac{-x - 2y}{3x^2 + y}$
9. The hyperbola $16x^2 - 9y^2 = 144$ has foci $S(5, 0)$ and $S'(-5, 0)$.
What are the equation of its' directrices?
- (A) $x = \frac{9}{5}$ and $x = -\frac{9}{5}$
- (B) $y = \frac{9}{5}$ and $y = -\frac{9}{5}$
- (C) $y = \frac{12}{5}$ and $y = -\frac{12}{5}$
- (D) $x = \frac{12}{5}$ and $x = -\frac{12}{5}$
10. A particle of mass m is projected vertically upwards with an initial velocity of $u \text{ ms}^{-1}$ in a medium in which the resistance to the motion is proportional to the square of the velocity $v \text{ ms}^{-1}$ of the particle or mkv^2 . Let x be the displacement in metres of the particle above the point of projection, O , so that the equation of motion is $\ddot{x} = -(g + kv^2)$ where $g \text{ ms}^{-2}$ is the acceleration due to gravity. Assume $k = 10$ and the acceleration due to gravity is 10 ms^{-2} .
Which of the following gives the correct expression for the time taken?
- (A) $t = \frac{1}{10}(\tan^{-1} u + \tan^{-1} v)$
- (B) $t = \frac{1}{10}(\tan^{-1} v - \tan^{-1} u)$
- (C) $t = \frac{1}{10}(\tan^{-1} u - \tan^{-1} v)$
- (D) $t = \frac{1}{10}(\tan^{-1} v + \tan^{-1} u)$

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet

Marks

- a)** Consider the complex numbers $\omega = -1 + \sqrt{3}i$ and $Z = \sqrt{3} + 2i$.
- | | | |
|-------|------------------------------------|---|
| (i) | Evaluate $\omega \bar{z}$. | 1 |
| (ii) | Evaluate $ \omega $. | 1 |
| (iii) | Find the value of $\arg(\omega)$. | 1 |
| (iv) | Find the value of ω^5 . | 1 |
| (v) | Evaluate $\frac{\omega}{Z}$. | 2 |
- b)** Sketch the region in the Argand diagram where $-\frac{\pi}{6} \leq \arg z \leq \frac{\pi}{3}$ and $z\bar{z} \leq 4$. 3
- c)** Evaluate $\int_0^4 x \sqrt{x^2 + 9} dx$. 3
- d)** Find $\int \frac{\sqrt{x^2 - 25}}{x} dx$, using the trigonometric substitution $x = 5 \sec \theta$. 3

Question 12 (15 marks) Use a SEPARATE writing booklet

Marks

- a)** A solid is formed by rotating about the y -axis the region bounded by the curve $y = \sin x$ and the x -axis between $0 \leq x \leq \pi$. Find the volume of this solid using the method of cylindrical shells. 4

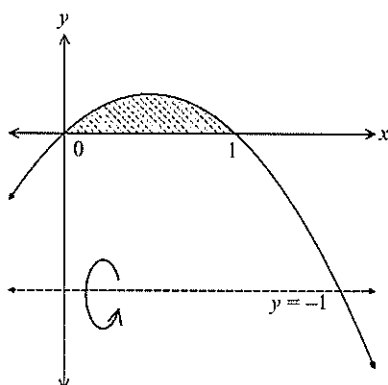
- b)** (i) If $\frac{x}{x^2 - x - 6} \equiv \frac{A}{x - 3} + \frac{B}{x + 2}$, find the values of A and B . 2

- (ii) Hence find $\int \frac{\sin \theta \cos \theta}{\sin^2 \theta - \sin \theta - 6} d\theta$ 2

- c)** Let two complex numbers be $z_1 = 2(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12})$ and $z_2 = 2i$.

- (i) On an Argand diagram sketch the vectors OA and OB to represent z_1 and z_2 respectively. 1
- (ii) Draw the vectors $z_1 + z_2$ and $z_1 - z_2$ on the same Argand diagram. 1
- (iii) What are the exact values of $\arg(z_1 + z_2)$ and $\arg(z_1 - z_2)$? 2

- d)** The area enclosed by the curve $y = x(1 - x)$ and the x -axis is rotated about the line $y = -1$. 3



Find the volume of the solid of revolution formed.

Question 13 (15 marks) Use a SEPARATE writing booklet.

Marks

- a)** A particle of mass m is moving in a straight line under the action of a force.

3

$$F = \frac{m}{x^3}(6 - 10x)$$

What is the velocity in any position, if the particle starts from rest at $x = 1$?

- b)** Consider the function $y = \cos^{-1}(e^x)$

(i) Find the domain and the range.

2

(ii) Sketch the graph of $y = \cos^{-1}(e^x)$?

2

(iii) Hence or otherwise sketch the graph of $y = [\cos^{-1}(e^x)]^2$.

1

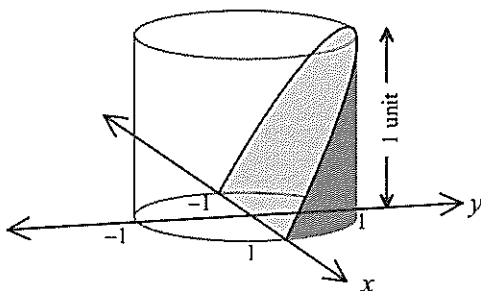
- c)** Use integration by parts to evaluate $\int_1^e \frac{\ln x}{x^2} dx$

3

- d)** A cylinder has the circle $x^2 + y^2 = 1$ as its base and is 1 unit in height.

4

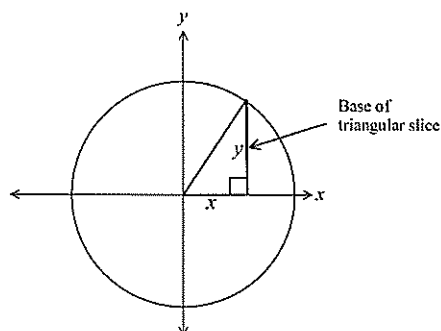
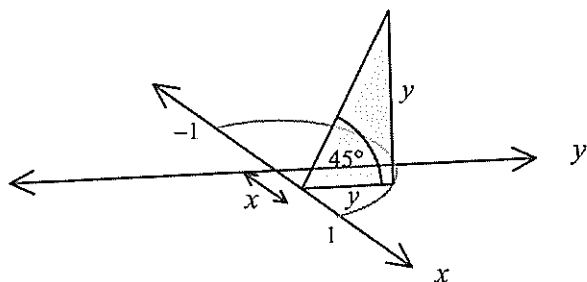
The shaded wedge is formed by a plane, which passes along the x -axis and is angled at 45° to the base of the cylinder.



Slices are taken through this wedge at right angles to the x axis, and perpendicular to the base of the cylinder, through a point (x, y) on the circle

Triangular slice through the wedge

Base of the cylinder

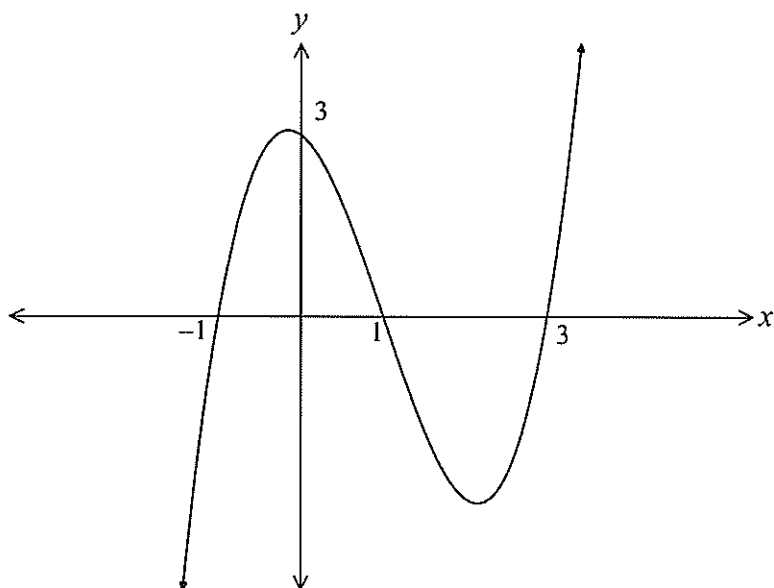


Find the volume of the wedge.

Question 14 (15 marks) Use a SEPARATE writing booklet.

Marks

- a)** A sketch of the function $f(x)$ is shown below.



Draw a separate half-page graph for each of the following functions, showing all asymptotes and intercepts.

- | | | |
|-------|----------------|---|
| (i) | $y = f(x) $ | 2 |
| (ii) | $y^2 = f(x)$ | 2 |
| (iii) | $y = f(x)$ | 2 |
| (iv) | $y = e^{f(x)}$ | 2 |

- b)** If one root of the equation $x^3 - px^2 + qx - r = 0$ is equal to the product of the other two, 3

show that:

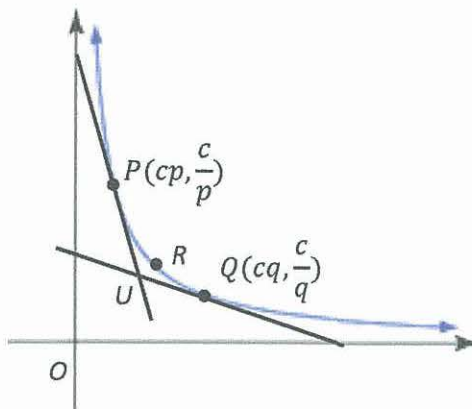
$$(q + r)^2 = r(p + 1)^2$$

- c)** Given that $P(x) = x^4 - 2x^3 + 2x - 1 = 0$ has a root of multiplicity 3, find the factors of $P(x)$. 4

Question 15 (15 marks) Use a SEPARATE writing booklet.

Marks

a)



On the hyperbola $xy = c^2$, three points P, Q and R are on the same branch, with parameters p, q and r respectively. The tangents at P and Q intersect in U .

- (i) If the equation of the tangent at P is $x + p^2y = 2cp$, show that the coordinates of U are: 1

$$\left(\frac{2cpq}{p+q}, \frac{2c}{p+q}\right)$$

- (ii) If O, U and R are collinear, prove that 3

$$r^2 = pq.$$

- b) (i) Let $I_n = \int_0^1 (1-x^r)^n dx$, where $r > 0$, for $n = 0, 1, 2, 3, \dots$ 3

$$\text{Show that } I_n = \frac{nr}{nr+1} I_{n-1}.$$

- (ii) Hence or otherwise, find the value of $I_n = \int_0^1 (1-x^{\frac{3}{2}})^3 dx$. 2

- c) (i) Show that the normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point $P(a \sec \theta, b \tan \theta)$ is given by: 3

$$ax \sin \theta + by = (a^2 + b^2) \tan \theta.$$

- (ii) If the normal meets the x -axis in G and PN is the perpendicular from P onto the x -axis, prove that $OG = e^2 ON$. 3

Question 16 (15 marks) Use a SEPARATE writing booklet. **Marks**

- a)** (i) Show that the condition for the line $y = mx + c$ to be a tangent to the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ is:
- $$c^2 = 16m^2 + 9$$
- (ii) Hence show that the pair of tangents drawn from (3, 4) to the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ are at right angles to each other. 3
- b)** A polynomial $P(x)$ is divided by $x^2 - a^2$, where $a \neq 0$, and the remainder is $px + q$.
- (i) Show that $p = \frac{1}{2a}\{P(a) - P(-a)\}$ and $q = \frac{1}{2}\{P(a) + P(-a)\}$. 2
- (ii) Find the remainder when the polynomial $P(x) = x^n - a^n$ is divided by $x^2 - a^2$ for the cases:
- (α) n even 1
- (β) n odd. 1
- c)** A mass of 1 kg is moving along the x –axis under the influence of two forces: an accelerating force of $\frac{F}{v}$ and a resisting force of kv^2 , where v is the velocity of the mass.
- (i) Write down the equation of motion. 1
- (ii) If the maximum velocity attained is V , show that $k = \frac{F}{V^3}$ 2
- (iii) Show that the distance travelled from $v = \frac{V}{4}$ to $v = \frac{V}{2}$ is $\frac{V^3}{3F} \ln \frac{9}{8}$. 3

Q1 $z = 2 - 3i$

$$\frac{1}{z} = \frac{1}{2-3i} \times \frac{2+3i}{2+3i}$$

$$= \frac{2+3i}{4+9}$$

$$= \frac{1}{13} (2+3i)$$

B.

Q2. $x = 4\sin^2\theta$ when $x=0, \theta=0$
 $\frac{dx}{d\theta} = 8\sin\theta\cos\theta$ $x=2, \theta=\pi/4$
 $dx = 8\sin\theta\cos\theta d\theta$

$$\int_0^2 \sqrt{\frac{x}{4-x}} dx = \int_0^{\pi/4} \sqrt{\frac{4\sin^2\theta}{4-4\sin^2\theta}} \cdot 8\sin\theta\cos\theta d\theta$$

$$= \int_0^{\pi/4} \sqrt{\frac{4\sin^2\theta}{4\cos^2\theta}} \cdot 8\sin\theta\cos\theta d\theta$$

$$= \int_0^{\pi/4} \frac{\sin\theta}{\cos\theta} \cdot 8\sin\theta\cos\theta d\theta$$

$$= \int_0^{\pi/4} 8\sin^2\theta d\theta$$

$$= 8 \int_0^{\pi/4} \frac{1}{2} (1 - \cos 2\theta) d\theta$$

$$= \frac{8}{2} \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\pi/4}$$

$$= \pi - 2$$

B.

Q3 For $x^3 + 3x^2 + 2x - 1 = 0$.

Let $y = \frac{2}{x} \therefore x = \frac{2}{y}$ sub $\frac{2}{y}$ for x

$$\left(\frac{2}{x}\right)^3 + 3\left(\frac{2}{x}\right)^2 + 2\left(\frac{2}{x}\right) - 1 = 0$$

$$\frac{8}{x^3} + 3 \times \frac{4}{x^2} + \frac{4}{x} - 1 = 0$$

mult by x^3

$$8 + 12x + 4x^2 - x^3 = 0$$

$$\therefore x^3 - 4x^2 - 12x - 8 = 0$$

A.

Q4 D.

D

Q5. Use the method of cylindrical shells

$$V = \int_a^b 2\pi xy \, dx$$

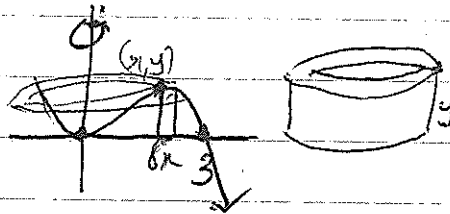
$$= \int_0^3 2\pi x (3x^2 - x^3) \, dx$$

$$= 2\pi \int_0^3 (3x^3 - x^4) \, dx$$

$$= 2\pi \left[\frac{3}{4} x^4 - \frac{x^5}{5} \right]_0^3$$

$$= 2\pi \left\{ \left[\frac{3}{4} (3^4 - \frac{3^5}{5}) \right] - \left[\frac{3}{4} 0 - \frac{0}{5} \right] \right\}$$

$$= \frac{243\pi}{10}$$



C

$$Q6 \quad z = 3\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$$

$$\bar{z} = 3\left(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6}\right)$$

$$\bar{z}^3 = 3^3 \left(\cos \frac{3\pi}{6} - i \sin \frac{3\pi}{6}\right)$$

$$= 27 \left(\cos \frac{\pi}{2} - i \sin \frac{\pi}{2}\right)$$

D

$$Q7 \quad \int_2^3 \frac{1}{\sqrt{4x-x^2}} dx$$

$$= \int_2^3 \frac{1}{\sqrt{-(x^2-4x+4)+4}} dx$$

$$= \int_2^3 \frac{1}{\sqrt{4-(x-2)^2}} dx$$

$$= \left[\sin^{-1} \left(\frac{x-2}{2} \right) \right]_2^3$$

$$= \sin^{-1} \frac{1}{2}$$

$$= \frac{\pi}{6}$$

C.

$$Q8 \quad x^3 + xy + y^2 = 7$$

By implicit differentiation

$$3x^2 \frac{dy}{dx} + y + 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} (x + 2y) = -3x^2 - y$$

$$\frac{dy}{dx} = \frac{-3x^2 - y}{x + 2y}$$

Gradient of tangent $m_1 = \frac{-3x^2 - y}{x + 2y}$

Gradient of normal $m_2 = \frac{x + 2y}{3x^2 + y}$

B.

Q9 $16x^2 - 9y^2 = 144$

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$

$$\therefore a=3, b=4$$

Foci are $(ae, 0)$ and $(-ae, 0)$
 $= (5, 0)$ and $(-5, 0)$

$$ae = 5$$

$$3e = 5$$

$$e = \frac{5}{3}$$

Equation of directrices

$$x = \pm \frac{a}{e}$$

$$= \pm \frac{3}{5/3}$$

$$= \pm \frac{9}{5}$$

A.

Q10 $\ddot{x} = -(g + kv^2)$
 $= -(10 + 10v^2)$

$$\frac{dv}{dt} = -10(1 + v^2)$$

$$\frac{dt}{dv} = \frac{-1}{10(1+v^2)}$$

$$t \Big|_0^v = \frac{-1}{10} \cdot \tan^{-1} v \Big|_u^v$$

$$t = -\frac{1}{10} [\tan^{-1} v - \tan^{-1} u]$$

$$= \frac{1}{10} [\tan^{-1} u - \tan^{-1} v]$$

C

Ext 2 Trial Exam 2016 St George Girls HS - Solutions

Solution

Mark

Q11

$$\begin{aligned} \text{a) i)} \quad w &= -1 + \sqrt{3}i & z &= \sqrt{3} + 2i & \bar{z} &= \sqrt{3} - 2i \\ w\bar{z} &= (-1 + \sqrt{3}i)(\sqrt{3} - 2i) \\ &= -\sqrt{3} + 2i + 3i + 2\sqrt{3} \\ &= \sqrt{3} + 5i \end{aligned}$$

1 mark for correct answer (1)

$$\begin{aligned} \text{ii)} \quad |w| &= \sqrt{(-1)^2 + (\sqrt{3})^2} \\ &= \sqrt{4} \\ &= 2 \end{aligned}$$

1 mark for correct answer (1)

$$\text{iii)} \quad \arg w = \tan^{-1}\left(\frac{\sqrt{3}}{-1}\right)$$

$= \theta$

$$\begin{aligned} \text{related angle} &= \tan^{-1}\sqrt{3} \\ &= \frac{\pi}{3} \end{aligned}$$

but w is in 2nd quadrant

$$\begin{aligned} \therefore \arg w &= \pi - \frac{\pi}{3} \\ &= \frac{2\pi}{3} \end{aligned}$$

1 mark

(1)

$$\text{iv)} \quad w = 2\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$$

$$\begin{aligned} w^5 &= 2^5 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)^5 \\ &= 32 \left(\cos \frac{10\pi}{3} + i \sin \frac{10\pi}{3}\right) \end{aligned}$$

$\left. \begin{array}{l} \frac{1}{2} \\ \frac{1}{2} \end{array} \right\}$

$$\begin{aligned} &= 32 \left(\cos -\frac{2\pi}{3} + i \sin -\frac{2\pi}{3}\right) \\ &= 32 \left(-\frac{1}{2} - i \frac{\sqrt{3}}{2}\right) = -16 - 16\sqrt{3}i \end{aligned}$$

$\frac{1}{2}$

(1)

$$\text{v)} \quad \frac{w}{z} = \frac{-1 + \sqrt{3}i}{\sqrt{3} + 2i} \times \frac{\sqrt{3} - 2i}{\sqrt{3} - 2i}$$

$$= \frac{\sqrt{3} + 5i}{3 + 4}$$

from (i)

1

$$= \frac{\sqrt{3} + 5i}{7}$$

1

(2)

Solutions

Mark

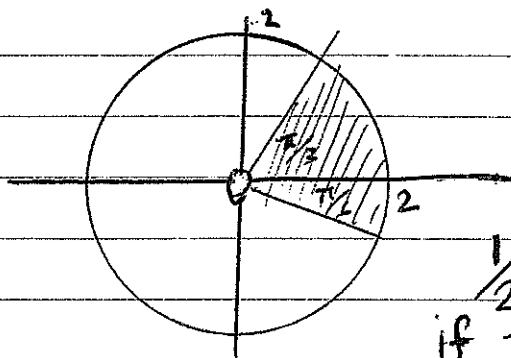
Q11

b) $z\bar{z} = 4$

$$(x-iy)(x+iy) = 4$$

$$x^2 + y^2 = 4$$

$z\bar{z} \leq 4$ is interior of circle



$\frac{1}{2}$ mark
if the origin
inclusive.

1 mark for
drawing correct circle

1 mark for correct range
of argument.

1 mark for correct
region.

c) $I = \int_0^4 x \sqrt{x^2 + 9} dx$

Let $u = x^2 + 9$

$$du = 2x dx$$

when $x=0$, $u=9$

$x=4$, $u=25$

$$\therefore I = \frac{1}{2} \int_9^{25} \sqrt{u} du$$

$$= \frac{1}{2} \int_9^{25} \sqrt{u} du$$

$$= \frac{1}{2} \left[\frac{2}{3} u^{\frac{3}{2}} \right]_9^{25}$$

$$= \frac{1}{3} [25^{\frac{3}{2}} - 9^{\frac{3}{2}}]$$

$$= \frac{1}{3} [125 - 27]$$

$$= \frac{98}{3} \quad \text{or} \quad 32 \frac{2}{3}$$

1 mark for substitution
and change of
limits.

1 mark for
integrand

1 mark for
answer.

③

③

Solution

Mark

$$Q11d) I = \int \frac{\sqrt{x^2-25}}{x} dx$$

$$\text{Let } x = 5 \sec \theta$$

$$dx = 5 \sec \theta \tan \theta d\theta$$

$$I = \int \frac{\sqrt{(5 \sec \theta)^2 - 25}}{5 \sec \theta} \cdot 5 \sec \theta \tan \theta d\theta$$

1 mark

$$= \int \frac{\sqrt{25 \sec^2 \theta - 25}}{5 \sec \theta} \cdot 5 \sec \theta \tan \theta d\theta$$

$$= \int \sqrt{25(\sec^2 \theta - 1)} \tan \theta d\theta$$

$$= \int 5 \sqrt{\tan^2 \theta} \cdot \tan \theta d\theta$$

$$= 5 \int \tan \theta \cdot \tan \theta d\theta$$

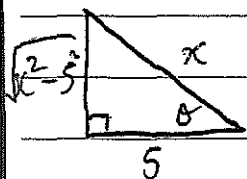
$$= 5 \int \tan^2 \theta d\theta$$

$$= 5 \int \sec^2 \theta - 1 d\theta$$

$$= 5 \tan \theta - 5\theta + c$$

1 mark

A few students did not change the variable back to x .



$$\sec \theta = \frac{x}{5}$$

$$\cos \theta = \frac{5}{x}$$

$$\therefore I = 5 \left(\frac{\sqrt{x^2-25}}{5} \right) - 5 \sec^{-1} \left(\frac{x}{5} \right) + c$$

$$= \sqrt{x^2-25} - 5 \sec^{-1} \left(\frac{x}{5} \right) + c$$

1 mark for correct answer

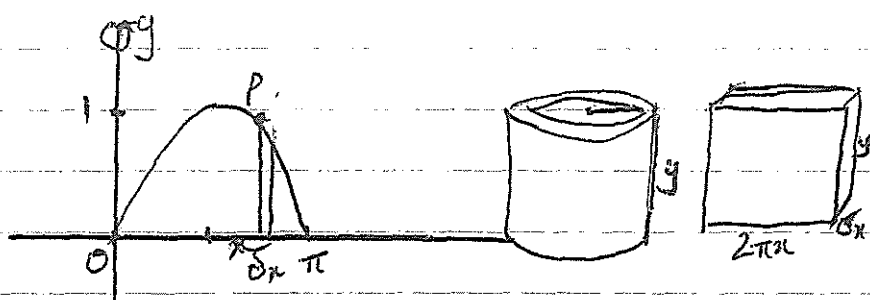
(3)

Solution

Mark

Q12

a)



1 mark

$$A = 2\pi xy$$

$$= 2\pi x \sin x$$

$$\delta V = 2\pi x \sin x \delta x$$

1 mark

Done very well
by most students

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^{\pi} 2\pi x \sin x \delta x$$

$$= 2\pi \int_0^{\pi} x \sin x dx$$

Using Integration by parts

1 mark

$$= 2\pi$$

$$u = x \quad v' = \sin x$$

$$u' = 1 \quad v = -\cos x$$

$$V = 2\pi \left[-x \cos x \right]_0^{\pi} - \int_0^{\pi} -\cos x dx$$

$$= 2\pi \left[(\pi - 0) + [\sin x]_0^{\pi} \right]$$

$$= 2\pi [\pi + (0 - 0)]$$

$$= 2\pi^2$$

1 mark

Solution

Mark

Q12

b) i) $x = A(x+2) + B(x-3)$

when $x = -2$

$$-2 = A(0) + -5B$$

$$B = \frac{2}{5}$$

when $x = 3$

$$3 = A(5) + B(0)$$

$$A = \frac{3}{5}$$

1 mark

1 mark

$$\therefore \frac{x}{x^2-x-6} = \frac{3}{5(x-3)} + \frac{2}{5(x+2)} \quad \text{--- (1)}$$

ii) $\int \frac{\sin \theta \cos \theta}{\sin^2 \theta - \sin \theta - 6} d\theta$

Let $u = \sin \theta$

$du = \cos \theta d\theta$

$$= \int \frac{u du}{u^2 - u - 6}$$

$$= \int \left(\frac{3}{5(u-3)} + \frac{2}{5(u+2)} \right) du \quad \text{from (1)}$$

$$= \frac{3}{5} \int \frac{1}{u-3} du + \frac{2}{5} \int \frac{1}{u+2} du$$

$$= \frac{3}{5} \ln(u-3) + \frac{2}{5} \ln(u+2) + C$$

$$= \frac{3}{5} \ln(\sin \theta - 3) + \frac{2}{5} \ln(\sin \theta + 2) + C$$

Majority of students did well on this question

1 mark

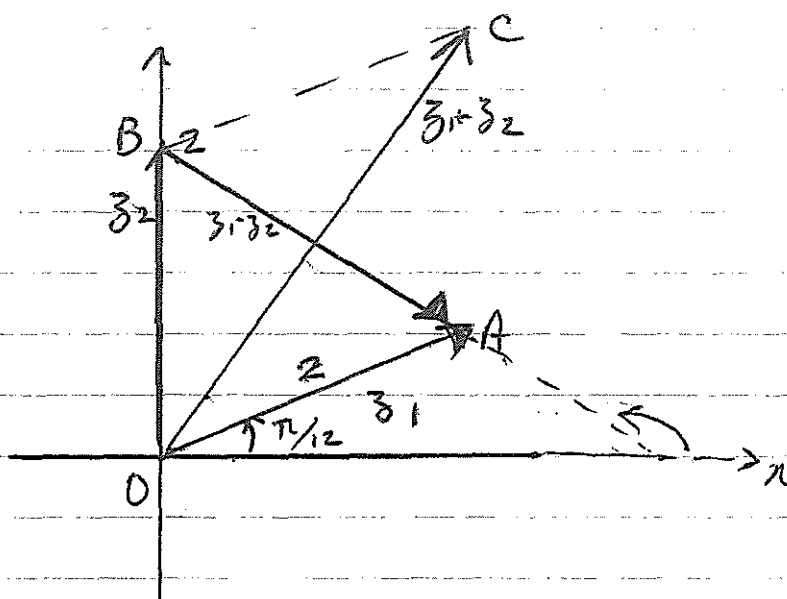
1 mark

Solution

Mark

Q12

c) i)



1 mark for
 z_1 at A and
 z_2 at B.

ii) On diagram

$\frac{1}{2}$ mark for vector
 $z_1 + z_2$

$\frac{1}{2}$ mark for vector

$$z_1 = z_2$$

iii) Let vector $z_1 + z_2$ be OC

Now OC and AB form a parallelogram.

Since $OA = OB = 2$

$\therefore$ OBCA is a rhombus.

Students had
problems with
finding the angle

$$\angle BOA = \frac{\pi}{2} - \frac{\pi}{12}$$

$$\therefore \angle AOC = \frac{1}{2} \times \left(\frac{\pi}{2} - \frac{\pi}{12} \right) \quad \text{diagonals bisect the angles through which they pass}$$

$$= \frac{5\pi}{24}$$

$$\therefore \arg(z_1 + z_2) = \frac{\pi}{12} + \frac{5\pi}{24}$$

$$= \frac{7\pi}{24}$$

1 mark.

Now $OC \perp AB$ (diagonals of a rhombus intersect at right angles)

$$\arg(z_1 - z_2) = -\frac{\pi}{2} + \frac{7\pi}{24} \quad \text{(exterior angle)}$$

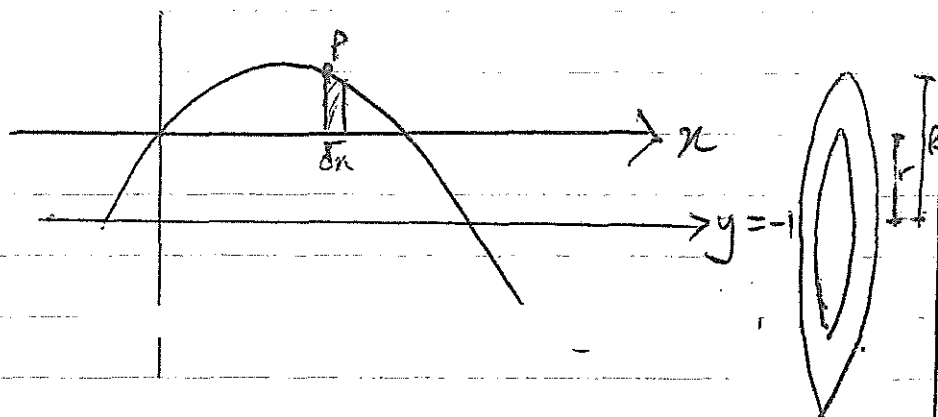
$$= -\frac{5\pi}{24}$$

1 mark

Solution

Mark

Q12
d) i)



$$A = \pi(R^2 - r^2)$$

$$R = y + 1$$

$$r = 1$$

$$y = x(1 - x)$$

$$= \pi((y+1)^2 - 1^2)$$

$$= \pi[(x - x^2 + 1)^2 - 1]$$

$$= \pi[(x - x^2)^2 + 2(x - x^2) + 1 - 1]$$

$$= \pi[x^2 - 2x^3 + x^4 + 2x - 2x^2]$$

$$\delta V = \pi[x^4 - 2x^3 - x^2 + 2x] \delta x$$

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^1 \pi[x^4 - 2x^3 - x^2 + 2x] \delta x$$

$$= \pi \int_0^1 (x^4 - 2x^3 - x^2 + 2x) dx$$

$$= \pi \left[\frac{x^5}{5} - \frac{2x^4}{2} - \frac{x^3}{3} + x^2 \right]_0^1$$

$$= \pi \left[\left(\frac{1}{5} - \frac{1}{2} - \frac{1}{3} + 1 \right) - 0 \right]$$

$$V = \frac{11\pi}{30} \text{ cu.}$$

Students did not interpret the line about which the solid is rotated properly and so got the radii incorrect

} 1

1

MATHEMATICS EXTENSION 2- QUESTION 13

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

$$13 a) \quad F = \frac{m}{x^3} (6 - 10x)$$

$$\text{If } F = ma$$

$$ma = \frac{m}{x^3} (6 - 10x)$$

$$a = \frac{6}{x^3} - \frac{10}{x^2}$$

$$\frac{d(\frac{1}{2}v^2)}{dx} = 6x^{-3} - 10x^{-2}$$

$$\frac{1}{2}v^2 = \int 6x^{-3} - 10x^{-2} dx$$

$$= -3x^{-2} + 10x^{-1} + c$$

$$\text{when } v = 0, \quad x = 1$$

$$\frac{1}{2}(0) = -3 + 10 + c$$

$$c = -7$$

$$\therefore \frac{1}{2}v^2 = \frac{-3}{x^2} + \frac{10}{x} - 7$$

$$v^2 = \frac{-6}{x^2} + \frac{20}{x} - 14$$

$$v = \pm \sqrt{\frac{-6}{x^2} + \frac{20}{x} - 14}$$

$$= \pm \sqrt{\frac{-6 + 20x - 14x^2}{x^2}}$$

$$v = \pm \frac{1}{x} \sqrt{2(-3 + 10x - 7x^2)}$$

Students who did not use

$$v' = \frac{d(\frac{1}{2}v^2)}{dx}$$

1 mark

found the question more difficult.

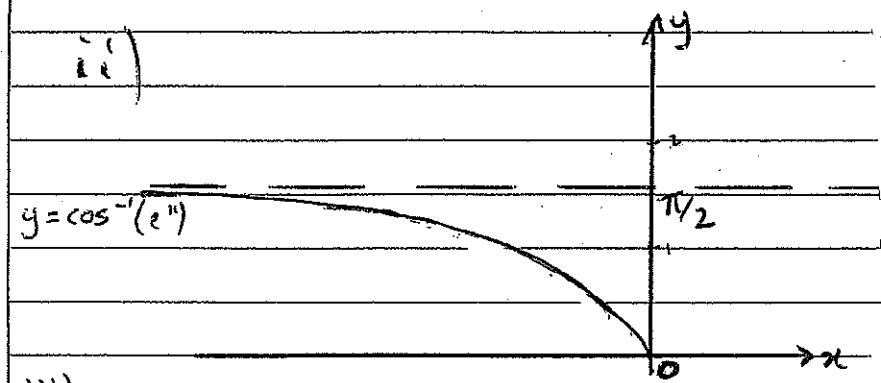
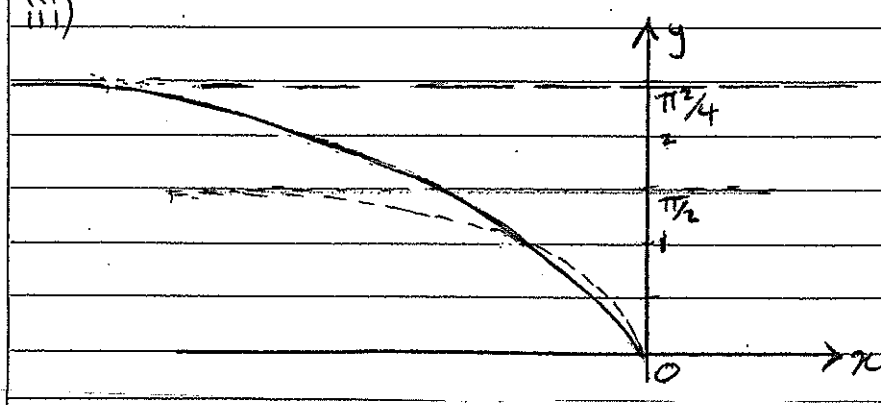
1 mark

1 mark

1/2 mark off if v was not $\pm$.

3

MATHEMATICS EXTENSION 2 - QUESTION 13

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
13b) i) For $y = \cos^{-1}x$ $D_f: -1 < x < 1$ $\therefore D_f: -1 \leq e^x \leq 1$ but $e^x > 0$ for all $x \therefore 0 < e^x \leq 1$ For $y = \cos^{-1}(e^x)$ $D: x \leq 0$ $R: 0 \leq y < \frac{\pi}{2}$	1 1	Many students included $\frac{\pi}{2}$ in the range but in the diagram had it as an asymptote.
ii)  iii) 	1 mark 1 mark	This graph was mostly drawn correctly, however the graphs are still <u>not</u> drawn $\frac{1}{3}$ page This question was poorly done. Many students did not draw the original graph correctly which therefore produced a change in concavity at the point where the graph met $y=1$.
Note: $\frac{1}{2}$ mark off if the graph showed any change of concavity		

MATHEMATICS EXTENSION 2 - QUESTION 13

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

$$c) \text{ Let } I = \int_1^e \frac{\ln x}{x^2} dx$$

$$\text{Let } u = \ln x \quad v' = x^{-2}$$

$$u' = \frac{1}{x} \quad v = -\frac{1}{x}$$

This question was done well overall.

$$\therefore I = \left[-\frac{\ln x}{x} \right]_1^e - \int_1^e -\frac{1}{x^2} dx$$

1

$$= \left[\left(-\frac{\ln e}{e} - 0 \right) + \int_1^e x^{-2} dx \right]$$

1

$$= -\frac{1}{e} - \left[x^{-1} \right]_1^e$$

$$= -\frac{1}{e} - \left(\frac{1}{e} - 1 \right)$$

$$= 1 - \frac{2}{e}$$

1

Solution

Mark

Q13 d)

If $x^2 + y^2 = 1$
Semi-circle: $y = \sqrt{1 - x^2}$

Area of triangle:

$$A(x) = \frac{1}{2} b h$$

$$= \frac{1}{2} y^2$$

$$= \frac{1}{2} (1 - x^2)$$

$$V = \int_0^1 A(x) dx$$

$$= \int_{-1}^1 \frac{1}{2} (1 - x^2) dx$$

$$= 2 \int_0^1 \frac{1}{2} (1 - x^2) dx$$

$$= \int_0^1 1 - x^2 dx$$

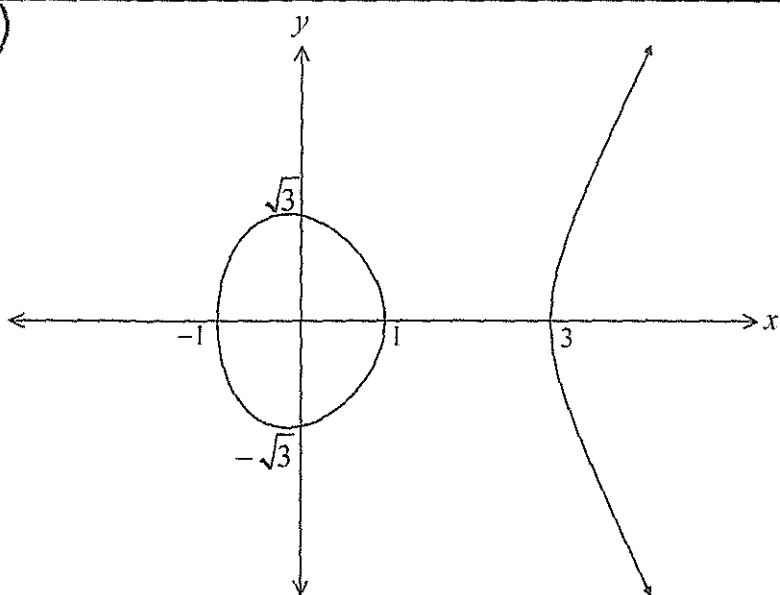
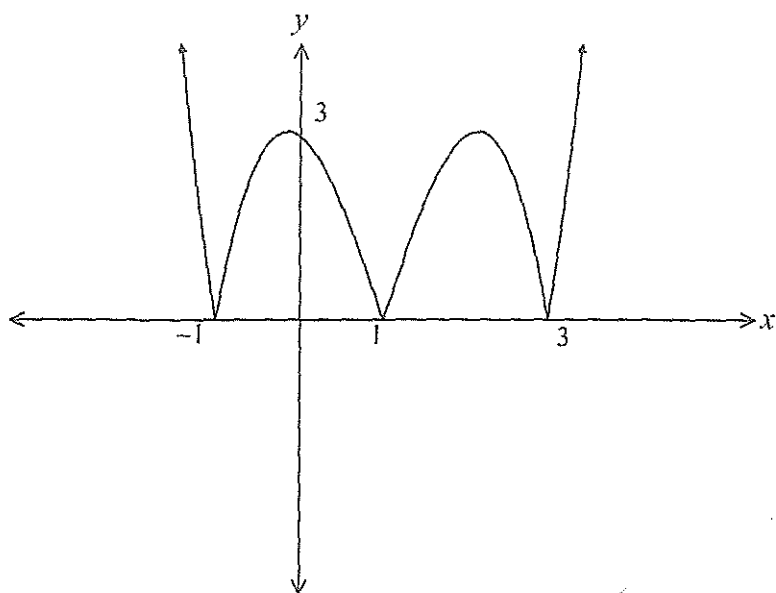
$$= \left[x - \frac{x^3}{3} \right]_0^1$$

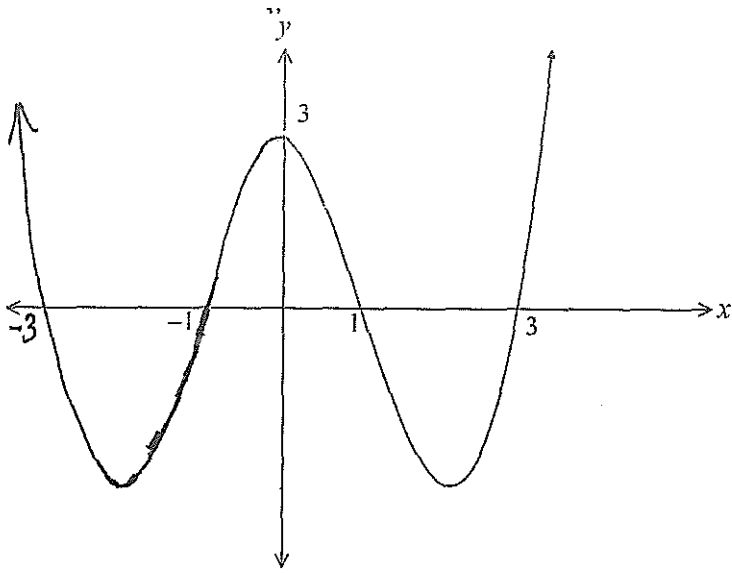
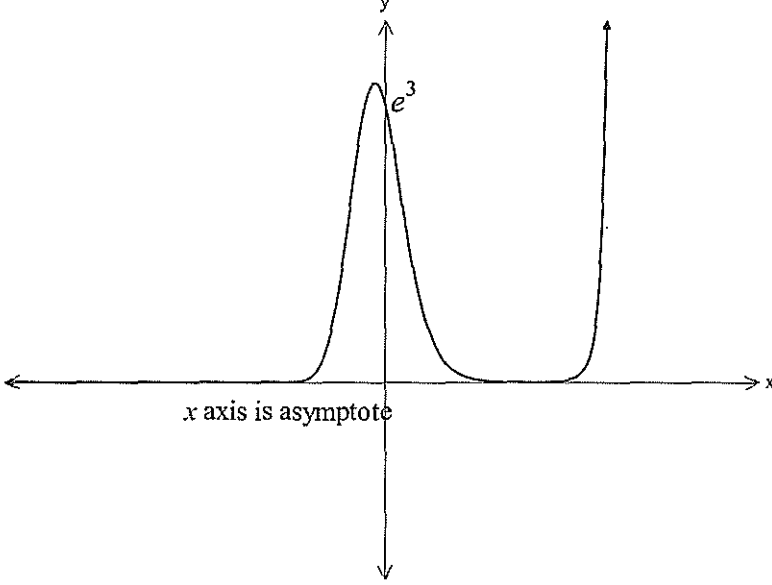
$$= 1 - \frac{1}{3}$$

$$= \frac{2}{3} u^2$$

(4)

Question 14		2016	
Solution		Marks	Allocation of marks
a)	(i)	2	2 marks for a correct graph with intercepts shown. 1 mark for graph with wrong intercepts or wrong orientation, or other minor error. <i>Very well done</i>
	(ii)		2 marks for a correct graph with intercepts shown. 1 mark for graph with wrong intercepts or wrong orientation, or other minor error. <i>Well done except for as $x \rightarrow \infty$</i>



Question 14		2016	
	Solution	Marks	Allocation of marks
	(iii) 	2	2 marks for a correct graph with intercepts shown. 1 mark for graph with wrong intercepts or wrong orientation, or other minor error. <i>Well done</i>
	(iv) 	2	2 marks for a correct graph with intercepts shown. 1 mark for graph with wrong intercepts or wrong orientation, or other minor error. <i>Most had problems with the local maximum</i>

Solution

Mark

Q14

b) Let roots be $\alpha, \beta, \alpha\beta$

$$\Sigma \alpha: \alpha + \beta + \alpha\beta = p \quad \dots \textcircled{1}$$

$$\Sigma \alpha\beta: \alpha\beta + \alpha^2\beta + \beta^2\alpha = q \quad \dots \textcircled{2}$$

$$\Sigma \alpha\beta\gamma: \alpha^2\beta^2 = r \quad \dots \textcircled{3}$$

$\textcircled{2} + \textcircled{3}$

$$\begin{aligned} (q+r) &= \alpha^2\beta^2 + \alpha\beta + \alpha^2\beta + \beta^2\alpha \\ &= \alpha\beta(1 + \alpha\beta + \alpha + \beta) \\ &= \alpha\beta(1 + p) \quad \text{from } \textcircled{1} \end{aligned}$$

$$(q+r)^2 = \alpha^2\beta^2(1+p)^2$$

$$(q+r)^2 = r(1+p)^2 \quad \#$$

$\textcircled{3}$

c) Let $P(x) = x^4 - 2x^3 + 2x - 1$

$$P'(x) = 4x^3 - 6x^2 + 2$$

$$\begin{aligned} P''(x) &= 12x^2 - 12x \\ &= 12x(x-1) \end{aligned}$$

When $x=0$ $P(0) \neq P'(0) \neq P''(0) \neq 0$

When $x=1$ $P(1) = 1 - 2 + 2 - 1 = 0$

$$\begin{aligned} P'(1) &= 4 - 6 + 2 \\ &= 0 \end{aligned}$$

$$P''(1) = 0$$

Since $P(1) = P'(1) = P''(1) = 0$

$\therefore (x-1)$ is a factor of multiplicity 3

$$\begin{aligned} \therefore P(x) &= (x-1)^3 Q(x) \\ &= (x-1)^3 (ax+b) \end{aligned}$$

but $a=1$ monic polynomial
 $b=1$

$$\therefore P(x) = (x-1)^3(x+1)$$

Well done by most students

Well done by most students

Some students did not write as factors

$\textcircled{4}$

Solution

Mark

Q15

a) i) The equation of the tangent at P is
 $x + p^2 y = 2cp \quad \text{--- (1)}$

This question was done very well

The equation of the tangent at Q is

$$x + q^2 y = 2cq \quad \text{--- (2)}$$

Solving simultaneously

$$(1) - (2)$$

$$y(p^2 - q^2) = 2c(p - q)$$

$$y(p - q)(p + q) = 2c(p - q)$$

$$y = \frac{2c}{p + q}$$

1/2

Sub in (1)

$$x + p^2 \left(\frac{2c}{p + q} \right) = 2cp$$

$$x = 2cp - \frac{2cp^2}{p + q}$$

1/2

$$= \frac{2cp(p + q) - 2cp^2}{p + q}$$

$$= \frac{2cp^2 + 2cpq - 2cp^2}{p + q}$$

$$= \frac{2cpq}{p + q}$$

$$\therefore u = \left(\frac{2cpq}{p + q}, \frac{2c}{p + q} \right)$$

(1)

MATHEMATICS EXTENSION 2- QUESTION 15

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
15a) i) Method 1: Gradient of OU, $m_{OU} = \frac{2c}{p+q}$ $\frac{\frac{2cpq}{p+q}}{\frac{2cpq}{p+q}}$ $= \frac{2c}{p+q} \times \frac{p+q}{2cpq}$ $= \frac{1}{pq}$ Equation of OU is: $y - 0 = \frac{1}{pq} (x - 0)$ $y = \frac{x}{pq} \quad \text{--- (3)}$ Now $xy = c^2$ --- (4) Sub (3) in (4) $x \left(\frac{x}{pq} \right) = c^2$ $\frac{x^2}{pq} = c^2$ $x^2 = c^2 pq \quad \text{--- (5)}$ but R is $(cr, \frac{c}{r})$ Sub $x = cr$ in (5) $(cr)^2 = c^2 pq$ $\cancel{c^2} r^2 = \cancel{c^2} pq$ $r^2 = pq$ Method 2: If O, U, R is collinear $m_{OU} = m_{OR}$ $\frac{\frac{2c}{p+q} - 0}{\frac{2cpq}{p+q} - 0} = \frac{\frac{c}{r} - 0}{cr - 0}$ $\frac{2}{2pq} = \frac{1}{r}$ $\frac{1}{pq} = \frac{1}{r^2}$ $r^2 = pq$	1 1 1 3 1/2 1	This question was mostly done well. If finding the equation of the line, OU, it is best to use the origin as the point not U. Some Students who used this method, did not use the origin as the common point. Instead they used U. This made the question more difficult i.e. $m_{UR} = m_{OU}$

Solution

Marks

Q15

$$b) I_n = \int_0^1 (1-x^r)^n dx$$

Using Integration by parts

$$u = (1-x^r)^n \quad v' = 1$$

$$u' = n(1-x^r)^{n-1} \cdot -rx^{r-1} \quad v = x$$

$$I_n = \left[x(1-x^r)^n \right]_0^1 - n \int_0^1 x(1-x^r)^{n-1} \cdot -rx^{r-1} dx$$

$$= 0 - nr \int_0^1 x(1-x^r)^{n-1} \cdot -x^{r-1} dx$$

$$= -nr \int_0^1 -x^r (1-x^r)^{n-1} dx$$

$$= -nr \int_0^1 [(1-x^r) - 1] (1-x^r)^{n-1} dx$$

$$= -nr \left(\int_0^1 (1-x^r)^n - (1-x^r)^{n-1} dx \right)$$

$$I_n = -nr (I_n - I_{n-1}) = -nr I_n + nr I_{n-1}$$

$$I_n(nr+1) = nr I_{n-1}$$

$$I_n = \frac{nr}{nr+1} \cdot I_{n-1}$$

$$ii) r = \frac{3}{2} \quad n = 3$$

$$I_3 = \frac{3 \times \frac{3}{2}}{3 \times \frac{3}{2} + 1} \times I_2 = \frac{9}{11} I_2$$

$$I_2 = \frac{2 \times \frac{3}{2}}{2 \times \frac{3}{2} + 1} \times I_1 = \frac{3}{4} I_1$$

$$I_1 = \frac{1 \times \frac{3}{2}}{1 \times \frac{3}{2} + 1} \times I_0 = \frac{3}{5} I_0 \quad I_0 = \int_0^1 1 dx = 1$$

$$I_3 = \frac{9}{11} \times \frac{3}{4} \times \frac{3}{5} \times 1$$

$$= \frac{81}{220}$$

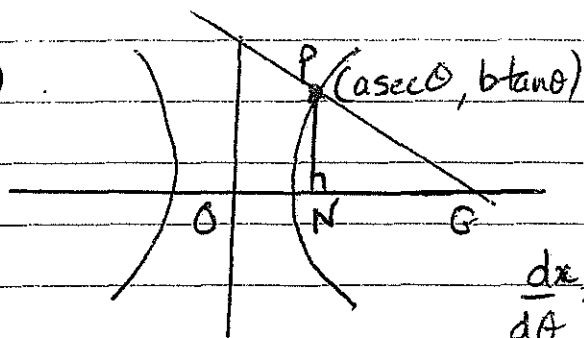
(2)

Solution

Marks

Q15

c) i)



$$x = a \sec \theta \quad y = b \tan \theta$$

$$\frac{dx}{d\theta} = a \sec \theta \tan \theta \quad \frac{dy}{d\theta} = b \sec^2 \theta$$

$$\therefore \frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx}$$

$$= \frac{b \sec^2 \theta}{a \sec \theta \tan \theta}$$

$$m_T = \frac{b \sec \theta}{a \tan \theta}$$

$$m_N = \frac{-a \tan \theta}{b \sec \theta} = -\frac{a \sin \theta}{b}$$

Equation of normal :

$$y - b \tan \theta = \frac{-a \tan \theta}{b \sec \theta} (x - a \sec \theta)$$

$$yb \sec \theta - b^2 \tan \theta \sec \theta = -ax \tan \theta + a^2 \tan \theta \sec \theta$$

$$yb \sec \theta + ax \tan \theta = (a^2 + b^2) (\tan \theta \sec \theta)$$

$$\frac{y}{\sec \theta} + ax \frac{\sin \theta}{\cos \theta} = (a^2 + b^2) \tan \theta$$

$$yb + ax \sin \theta = (a^2 + b^2) \tan \theta$$

ii) For G, when $y=0$, $x = \frac{(a^2 + b^2) \tan \theta}{a \sin \theta}$

From $b^2 = a^2(e^2 - 1)$

$$b^2 = a^2 e^2 - a^2$$

$$a^2 + b^2 = a^2 e^2$$

and $ON = x_N = a \sec \theta$

$$x_G = \frac{e^2 a^2}{a \cos \theta}$$

$$= e^2 a \sec \theta$$

$$OG = e^2 \cdot ON$$

(3)

Solution

Mark

Q16

a) i) Solve $y = mx + c$ and $\frac{x^2}{16} + \frac{y^2}{9} = 1$

simultaneously:

$$\frac{x^2}{16} + \frac{(mx+c)^2}{9} = 1$$

$$9x^2 + 16(m^2x^2 + 2mxc + c^2) = 144$$

$$9x^2 + 16m^2x^2 + 32mcx + 16c^2 = 144$$

$$x^2(9 + 16m^2) + 32mcx + 16c^2 - 144 = 0$$

For the line to be a tangent, $\Delta = 0$

$$32^2 m^2 c^2 - 4(9 + 16m^2)(16c^2 - 144) = 0$$

$$1024m^2c^2 - 4(144c^2 - 1296 + 256m^2c^2 - 2304) = 0$$

$$1024m^2c^2 - 576c^2 + 5184 - 1024m^2c^2 + 9216m^2 = 0$$

$$576c^2 = 9216m^2 + 5184$$

$$\div 576 \quad c^2 = 16m^2 + 9 \quad \#$$

ii) Let the tangents have the equation

$$y = mx + c \quad \text{so at } (3, 4)$$

$$4 = 3m + c$$

$$c = 4 - 3m \quad \text{--- (1)}$$

Condition for tangents is $c^2 = 16m^2 + 9$ --- (2)

Sub (1) into (2)

$$(4 - 3m)^2 = 16m^2 + 9$$

$$16 - 24m + 9m^2 = 16m^2 + 9$$

$$7m^2 + 24m - 7 = 0$$

Roots of this quadratic equation in m are gradients of the two tangents. So

$$\text{Product of roots} = \frac{c}{a}$$

$$= \frac{-7}{7}$$

$$= -1$$

$\therefore$ tangents are perpendicular

1

Some students differentiated but were able to

} 1 change to this

} 1

(3)

Some students had problems forming the equation in m and c .

1

Most student found the product of the gradients rather than the product of the roots (2)

Solution

Q16

b) i) $P(x) = (x^2 - a^2)Q(x) + (px + q)$

$$P(a) = pa + q \quad \text{--- (1)}$$

$$P(-a) = -pa + q \quad \text{--- (2)}$$

Well done by most students

(1) - (2)

$$P(a) - P(-a) = 2 \times pa$$

$$\therefore p = \frac{1}{2a} (P(a) - P(-a))$$

(1) + (2)

$$P(a) + P(-a) = 2q$$

$$\therefore q = \frac{1}{2} (P(a) + P(-a))$$

ii) if $P(x) = x^n - a^n$

(A) when n is even then $P(a) = 0$ & $P(-a) = 0$

$$\therefore p = 0, q = 0 \therefore px + q = 0$$

(B) when n is odd then

$$P(a) = 0 \text{ and } P(-a) = -2a^n$$

$$\therefore p = \frac{1}{2a} (0 + 2a^n) \quad q = \frac{1}{2} (0 - 2a^n)$$

$$R = px + q$$

$$= \frac{2a^n}{2a} x - a^n$$

$$R = a^{n-1} x - a^n$$

Many students had difficulty with this question

Solution

Marks

Q16

c) i) $\ddot{x} = \frac{F}{v} - kv^2$

1 mark

ii) For maximum velocity $\ddot{x} = 0$

$$0 = \frac{F}{v} - kv^2$$

$$kv^3 = F$$

$$k = \frac{F}{V^3}$$

1 mark

1 mark

(2)

Most students did not understand that F in the question was a constant

iii) $\dot{x} = v \cdot \frac{dv}{dx} = \frac{F}{v} - kv^2$

$$\frac{dv}{dx} = \frac{F}{v^2} - kv$$

$$= \frac{F - kv^3}{v^2}$$

1 mark

A lot of students used $F = ma$ or $F = m\ddot{x}$ but F was a constant

$$\frac{dx}{dv} = \frac{v^2}{F - kv^3}$$

$$x = \int_{V/4}^{V/2} \frac{v^2}{F - kv^3} dv$$

$$= -\frac{1}{3k} \int_{V/4}^{V/2} \frac{-3kv^2}{F - kv^3} dv$$

$$= -\frac{1}{3k} \left[\ln(F - kv^3) \right]_{V/4}^{V/2}$$

$$= -\frac{1}{3} \cdot \frac{1}{F/V^3} \left[\ln\left(F - k \cdot \frac{V^3}{8}\right) - \ln\left(F - k \frac{V^3}{64}\right) \right]$$

$$= -\frac{V^3}{3F} \ln \left[\frac{F - \frac{kV^3}{8}}{F - \frac{kV^3}{64}} \right]$$

1/2 mark

$$= \frac{V^3}{3F} \ln \left[\frac{F - \frac{kV^3}{64}}{F - \frac{kV^3}{8}} \right]$$

$$= \frac{V^3}{3F} \ln \left[\frac{64kV^3 - kV^3}{8kV^3 - kV^3} \right]$$

1/2 mark

$$= \frac{V^3}{3F} \ln \frac{63kV^3}{8 \times 7kV^3} = \frac{V^3}{3F} \ln \frac{9}{8}$$